

Application of ohmic heating on the bell pepper puree with added apple pectin



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Ohmic heating (OH) is a potential novel technology which can concur to improve the quality of food. The effects of OH at 20 V/cm applied for 3 min on the bell pepper purees with added apple pectin (0.1...0.3%) were evaluated. The quantification of the effects of OH over the products was carried out in terms of chemical properties, antioxidant capacity and total polyphenols determination, viscosity and texture analysis, Fourier transform infrared (FT-IR) spectroscopy and confocal scanning microscopy. The pH values after 30 days of storage varies from 4.22...4.45 with a decreasing of 5...15% reported to the blank sample. The inhibition of ABTS for all samples ranged from 45.85...54.47%, which indicates a very good antioxidant capacity. The sample with 0.3% apple pectin registered the highest value of total polyphenols (85.2 mg GAE/ml sample). The dynamic viscosity values of the samples versus share rate are arranged according to the Ostwald de Waele model, specific to rheological behavior of purees. The (FT-IR) spectroscopy and confocal scanning microscopy results indicate minimal changes in bell pepper purees with added apple pectin, recommending the OH as a mild processing.

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Amaranth – A promising crop for fodder manufacturing



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Amaranth is an annual plant of the amaranth family (Amaranthaceae), which, according to N.I. Vavilov, is destined to feed humanity. By its properties, it is unique fodder crop, the green mass yield of which makes up more than 50 t/ha. The fiber content in amaranth is 16–20%, the concentration of water-soluble sugars is 6.4–7.2%. Amaranth protein contains 6–9% of amino acid lysine, which is much higher than in the protein of maize, wheat or rice. The use of amaranth for the production of silage will enable to significantly facilitate the solution of the protein problem in cattle breeding. However, number of tasks related to its ensilaging have not been resolved: there is no consensus on the timing of amaranth harvesting for silage, its ensilage capacity has not been studied, there are no data on the preservation and quality of the fodder obtained as well as on the effectiveness of its use in cattle diet.

The authors have selected variants of starter cultures from several kinds of lactic acid bacteria for this hard-to-ensilage raw

material. The optimal proportion of cultures in starter formulations was determined, which makes it possible to obtain high quality silage from amaranth.

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Antimicrobial activity of essential oils of *Laurus nobilis*, *Rosmarinus officinalis*, and *Myrtus communis* on common foodborne pathogens



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Aromatic plants have been used in food, medicine, cosmetics as well as pharmaceutical industry for centuries. Essential oils have been great biotechnological potential due to their importance functions such as they effectively destroy microorganisms. Essential oils consist of chemical components such as terpenoids including monoterpenes, sesquiterpenes and their oxygenated derivatives. Antimicrobial activity of essential oils of *Laurus nobilis*, *Rosmarinus officinalis*, and *Myrtus communis* on *Escherichia coli* O157:H7, *Listeria monocytogenes*, *Salmonella Typhimurium*, *Salmonella Enteritidis*, and *Staphylococcus aureus* were determined by disk diffusion method. Air-dried leaves of the plants were steam-distilled using a Clevenger-type apparatus. A 100 µl portion of bacterial suspension was adjusted to 10⁸ CFU/ml and was spreaded on the surfaces of Mueller–Hinton (MH) agar plates. Paper disks were impregnated with 5 µl of essential oils and were placed on the surface of MH agar, then were incubated overnight at 37 °C. After 24 h, the diameter of the resulting zone of inhibition was measured. The obtained results showed the essential oils had a remarkable antimicrobial effect on the different bacterial strains as food additives to extend shelf life of foods. Moreover, *Laurus nobilis* and *Rosmarinus officinalis* showed the highest antibacterial activity against Gram-negative bacteria.

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Probiotic and functional characterization of *Lactobacillus* spp. isolated from infant faeces



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In the early stages of life, balanced colonization of the gastrointestinal system is crucial for acquiring immunity in the infancy period. *Lactobacillus* spp. is beneficial in human diet due to their potential therapeutic and probiotic properties. In this study, 104 *Lactobacillus* species isolated from faeces of babies and their probiotic and functional properties were investigated.

Faecal samples were collected from 14 healthy infants, which were fed with breast milk and their age between 3 and 47 weeks. Strains were identified by 16s rDNA sequencing and phenotypically characterized by testing their growth in MRS broth at 15 and 45 °C, production of CO₂ (gas production) from glucose, hydrolysis of arginine, and salt tolerance.

Isolated strains were identified as *L. paracasei* subsp. *paracasei* (41 strains), *L. fermentum* (24 strains), *L. rhamnosus* (11 strains), *L. casei* (17 strains), *Lactobacillus* spp. (11 strains) based on 16S rDNA sequence analysis. Almost all strains of *L. casei/L. paracasei* subsp. *paracasei*, *L. rhamnosus* and several strains of *L. fermentum* were able to produce hydrogen sulfide. Majority of isolates demonstrated an antibacterial activity against *E. coli* ATCC 25922, *E. coli* O157:H7, *L. monocytogenes*, *S. Enteritidis*, *S. aureus*. Our study demonstrated some important strains of Lactic acid bacteria isolated from baby faeces.

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Amino acid composition of gilthead sea bream fillets (*Sparus aurata*) coated with thymol-loaded chitosan nanofibers during cold storage



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Amino acid compositions of fish fillets may change during cold storage conditions. Therefore, an effort should also be exerted to prevent the change. Therefore, this study was undertaken to coat the fish fillets with chitosan nanofibers (CN) and thymol loaded chitosan nanofibers (TLCN) using electrospinning technique. Sixteen amino acids in fish fillets were tested in terms of storage stability at $4 \pm 1^\circ\text{C}$ in storage days (1st, 5th, 9th and 11th). As compared to those in control group (uncoated fish fillets), the amounts of lysine, phenylalanine, leucine, isoleucine, methionine, valine, tyrosine, proline, alanine, threonine, histidine, glycine, serine and glutamic acid were determined to be higher in fish fillets coated with TLCN. Total amounts of the amino acids values at 5th, 9th and 11th days of cold storage were determined as 3896, 6069 and 6277 mg/100 g in control fillets and the fillets coated with CN and TLCN, respectively. The results revealed that the fish fillets coated with TLCN samples were more stable in terms of amino acid composition as compared to uncoated fillets, suggesting that coating of fish fillets with nanofibers could be a novel approach in terms of stability of amino acid composition during cold storage.

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Fructans juice as fermentable raw material to improve a beverage with prebiotic and probiotic properties



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Nowadays, consumers are looking for variety and unconventional functional drinks. The taste grade is now huge and highly

dynamic because the young target group emphasizes miscellaneous and trendy flavorings. Despite a very pronounced premium and luxury image also this segment shows increased interest in a health positioning. Due to their beneficial effect on health fructans as prebiotic become more and more important in the Functional Food and Beverage Industry, fructose polymers are accumulated by plants such as *Cichorium* sp., *Allium* sp., *Asparagus* sp., and *Agave* sp. Therefore, the aim of this paper is to establish a feasible method to produce a fermented wine-type drink originated from the sugars of *Agave* sp. The brewing method was performed on the known manufacture process which is used to produce traditional grape wine with determined modification. Special consideration is focused on the fermentation process and essential microorganisms involved, their fermentation products, the biochemical changes of these unique fermentations, and their impact on the quality and sensorial characteristics of the product.

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Medical Genetics

Targeted custom gene panel sequencing for cardiac ion channelopathies: Efficiently detects candidate pathogenic mutations in Long QT syndrome



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Long QT syndrome is characterized by prolongation of the QT interval on electrocardiogram, and associated with a high risk of sudden death. To date mutations in 15 LQTS-susceptibility genes have been implicated. However, the genetic cause of approximately 20–25% of LQTS patients remains elusive. We screened for 68 cardiac arrhythmia genes among 92 unrelated LQT cases. Targeted sequencing on ion 'PGM' using Custom Panel revealed that known LQT gene variants accounted for just 40.5% of the cases. 168 missense, 11 splice site, 4 frameshifts, 6 in frame deletions, 2 nonsense and 4 'UTR' variants have been identified. 39 of these variants were novel, 26 of these variants were reported as pathogenic, 40 of these had